Analysis Questions for SQL Queries

In this project, we will focus on Business-to-Customer (B2C) sales of the AdventureWorksDW database. In other words, we will focus on internet sales and in the later projects, the business-to-business sales (reseller sales) will be analyzed, separately. The B2C - internet sales analysis will be done within the following sections:

1. Section 1: Executive Macro-Performance & Revenue Trends
2. Section 2: Customer Segmentation & Behavioral Profiling
3. Section 3: Product Affinity & Profitability Analysis
4. Section 4: Promotional ROI & Sales Drivers
5. Section 5: Geographic Distribution & Fulfillment Efficiency

The tables that are required for this project are `FactInternetSales`, `FactInternetSalesReason`, `Calendar`, `Customer`, `Geography`, `SalesTerritory`, `Product`, `ProductSubcategory`, `ProductCategory`, `SalesReason` and `Promotion`.

Section 1. Executive Macro-Performance & Revenue Trends

1. What is the overall Gross Profit and Gross Margin for the B2C channel, and how has it trended year over year?

- **Business Context:** Revenue alone is a vanity metric. The C-suite needs to know if the business is actually making money after the cost of goods sold (COGS).
- **The Task:** Calculate the Total Revenue, Total Cost, Gross Profit (Revenue - Cost), and Gross Profit Margin Percentage (Gross Profit / Revenue) for the entire dataset, broken down by `calendaryear`.
- **Tables & Columns:** * `factinternetsales`: `salesamount` (for Revenue), `totalproductcost` (for COGS).
  - `calendar`: `calendaryear`.
- **Technical Hint:** You will need basic aggregations like `sum` and math operations. Watch out for division by zero when calculating the margin, and use `cast` or `round` to format the percentages cleanly.

2. What is the Year-over-Year (YoY) revenue growth rate for each month?

- **Business Context:** Comparing January 2024 to January 2023 tells a much clearer story about growth than comparing January 2024 to December 2023 (due to holiday spikes).
- **The Task:** Calculate the total `salesamount` for each year and month. Then, compare each month's revenue to the exact same month in the previous year to find the percentage growth or decline.
- **Tables & Columns:** * `factinternetsales`: `salesamount`, `orderdatekey`.
  - `calendar`: `datekey`, `calendaryear`, `monthnumberofyear`, `monthname`.
- **Technical Hint:** This is a perfect opportunity to showcase window functions. You will want to use the `lag` function, utilizing `partition by monthnumberofyear` and `order by calendaryear` to fetch the previous year's revenue for the same month.

3. What is the monthly seasonality of our sales, and which quarter drives the most revenue historically?

- **Business Context:** Supply chain and marketing teams need to know when the busy seasons are so they can plan inventory and ad spend accordingly.
- **The Task:** Aggregate historical `salesamount` by `calendarquarter` and `monthname` across all years combined to identify the highest and lowest performing periods.
- **Tables & Columns:**
  - `factinternetsales`: `salesamount`, `orderdatekey`.
  - `calendar`: `datekey`, `monthname`, `monthnumberofyear` (for sorting), `calendarquarter`.
- **Technical Hint:** A simple `group by` is required here. Make sure to `order by` the `monthnumberofyear` rather than the alphabetical `monthname` so your results flow chronologically.

4. How is the Average Order Value (AOV) trending alongside Total Order Volume?

- **Business Context:** If revenue is growing, executives need to know *why*. Are we getting more customers to buy (Order Volume), or are the same number of customers just spending more money per transaction (AOV)?
- **The Task:** For each `calendaryear` and `monthnumberofyear`, calculate the total number of unique orders placed and the average revenue generated per order.
- **Tables & Columns:**
  - `factinternetsales`: `salesordernumber` (for counting orders), `salesamount`.
  - `calendar`: `calendaryear`, `monthnumberofyear`.
- **Technical Hint:** You must use `count` with `distinct` on `salesordernumber`, because one order can have multiple line items (rows) in the fact table. Divide the `sum` of `salesamount` by this distinct count to get the AOV.

5. What is the Year-to-Date (YTD) cumulative revenue for the most recent fully completed year?

- **Business Context:** Pacing is critical. Leaders look at running totals to see how quickly they are approaching annual financial targets compared to previous years.
- **The Task:** Filter the data for the most recent complete year. Calculate the running total of `salesamount` day by day.
- **Tables & Columns:**
  - `factinternetsales`: `salesamount`, `orderdatekey`.
  - `calendar`: `datekey`, `fulldatealternatekey`, `calendaryear`.
- **Technical Hint:** You will need an advanced window function here: `sum` used with an `over` clause, ordering by the date, to create a cumulative rolling sum.

Section 2: Customer Segmentation & Behavioral Profiling

6. Which demographic segments (based on Income and Education) drive the highest overall revenue?

- **Business Context:** Marketing teams have limited budgets. They need to know if they should be targeting high-income professionals with postgraduate degrees or middle-income customers with bachelor's degrees.
- **The Task:** Segment the total revenue and the total number of unique orders by the customer's education level and their yearly income bracket.

- **Tables & Columns:**
  - `factinternetsales`: `salesamount`, `salesordernumber` (for order count).
  - `customer`: `education`, `yearlyincome`.
- **Technical Hint:** You will use a standard `group by` on the demographic columns. Use `sum` for the revenue and `count distinct` for the orders. You can order the results by the highest revenue to immediately see the most valuable persona.

7. Does the customer's commute distance and vehicle ownership impact their lifetime value?

- **Business Context:** Because AdventureWorks sells bicycles, a senior analyst would hypothesize that people with longer commutes or fewer cars might spend more on cycling gear. We need to prove or disprove this.
- **The Task:** Calculate the total revenue and average revenue per customer, grouping them by their `commutedistance` and `numbercarsowned`.
- **Tables & Columns:**
  - `factinternetsales`: `salesamount`, `customerkey`.
  - `customer`: `commutedistance`, `numbercarsowned`, `customerkey`.
- **Technical Hint:** `join` the fact and customer tables on `customerkey`. Group the data by the commute and car columns. Use an aggregation to find the total revenue, and divide that by the `count distinct` of `customerkey` to find the average spend per person in each group.

8. Who are our "Whales" (Top 100 Most Valuable Customers), and what is their purchasing frequency?

- **Business Context:** The Pareto Principle (80/20 rule) often applies in retail—a small percentage of customers drives a large percentage of revenue. The business needs to identify these VIPs for exclusive loyalty programs.
- **The Task:** Identify the top 100 individual customers based on their total lifetime spend. For each of these customers, calculate how many separate orders they have placed.
- **Tables & Columns:**
  - `factinternetsales`: `salesamount`, `salesordernumber`, `customerkey`.
  - `customer`: `firstname`, `lastname`, `customerkey`.
- **Technical Hint:** You will need to `group by` the customer's name or key. Use `sum` for the sales and `count distinct` for the orders. To get only the top 100, use `select top 100` (in sql server) and `order by` the total sales amount in `desc` order.

9. What is our generational spend distribution (Age Brackets)?

- **Business Context:** Are we primarily selling to younger adults or middle-aged enthusiasts? Age strongly dictates which marketing channels (e.g., TikTok vs. Facebook) the company should use.
- **The Task:** Calculate the age of each customer at the time of their purchase. Then, group these ages into logical buckets (e.g., Under 35, 35-44, 45-54, 55+) and calculate the total revenue for each bucket.
- **Tables & Columns:**
  - `factinternetsales`: `salesamount`, `orderdate`.
  - `customer`: `birthdate`, `customerkey`.
- **Technical Hint:** This requires some data transformation. You can use the `datediff` function to find the difference in years between `birthdate` and `orderdate`. Then, wrap that logic inside a `case`

`when` statement to categorize the calculated ages into your distinct generational buckets.

10. What is our Customer Retention Rate (One-and-Done vs. Repeat Buyers)?

- **Business Context:** Acquiring a new customer costs significantly more than retaining an existing one. If 90% of customers only buy once and never return, the business has a "leaky bucket" problem.
- **The Task:** Determine the percentage of our total customer base that has placed only 1 order in their lifetime versus those who have placed 2 or more orders.
- **Tables & Columns:**
  - `factinternetsales`: `customerkey`, `salesordernumber`.
- **Technical Hint:** This is a classic two-step problem. First, write a `cte` (common table expression) or subquery that uses `count distinct` on `salesordernumber` grouped by `customerkey` to find out exactly how many orders each person made. In your main query, use a `case when` statement against that `cte` to group the customers into "1 Order" and "Multiple Orders", then `count` how many customers fall into each bucket.

Section 3: Product Affinity & Profitability Analysis

11. Which product categories and subcategories drive the highest profit margins, and which are drag on profitability?

- **Business Context:** Bikes might generate the most raw revenue, but accessories (like water bottles and helmets) often have significantly higher profit margins. The business needs to know where the actual profit is coming from to optimize catalog placement.
- **The Task:** Calculate the total revenue, total cost, gross profit, and gross profit margin percentage for each `productcategoryname` and `productsubcategoryname`.
- **Tables & Columns:**
  - `factinternetsales`: `salesamount`, `totalproductcost`.
  - `product`: `productkey`, `productsubcategorykey`.
  - `productsubcategory`: `productsubcategorykey`, `productcategorykey`, `productsubcategoryname`.
  - `productcategory`: `productcategorykey`, `productcategoryname`.
- **Technical Hint:** You will need to chain several `left join` statements together to connect the fact table all the way up to the category table. Then, `group by` the category and subcategory names, applying `sum` to your financial columns.

12. What is the disparity between our "High Volume" products and our "High Value" products?

- **Business Context:** A $5 tire patch kit might sell 10,000 units (High Volume), while a $3,000 mountain bike might only sell 50 units (High Value). Understanding this split dictates warehouse storage strategies.
- **The Task:** Rank the top 10 products by total `orderquantity` (volume) and separately rank the top 10 products by total `salesamount` (value).
- **Tables & Columns:**
  - `factinternetsales`: `orderquantity`, `salesamount`, `productkey`.

  - `product`: `productkey`, `productname`.
- **Technical Hint:** You can solve this effectively by writing two separate `cte` structures (one for volume, one for value) using the `select top 10` clause and `order by ... desc`. Alternatively, you can pull a larger list and use the `rank()` window function over both metrics to compare them side-by-side.

13. What is our basket size distribution, and how frequently do customers buy multiple items in a single transaction?

- **Business Context:** E-commerce platforms thrive on upselling. If the vast majority of our orders only contain a single item, our "frequently bought together" recommendation engine is failing.
- **The Task:** Determine the distribution of order sizes (e.g., how many orders contained exactly 1 item, 2 items, 3 items, etc.).
- **Tables & Columns:**
  - `factinternetsales`: `salesordernumber`, `orderquantity`.
- **Technical Hint:** This requires a subquery. First, `group by salesordernumber` and use `sum` on `orderquantity` to find the total items per cart. Then, in your outer query, `group by` that calculated total cart size and `count` how many distinct `salesordernumber` fall into each bucket.

14. Are there any "Dead Weight" products in our active catalog that have negligible sales in the most recent year?

- **Business Context:** Warehousing unsold goods costs money. If an active product isn't selling, the business needs to discount it heavily to clear it out and discontinue it.
- **The Task:** Focusing only on the most recent full calendar year of data, identify any products that are marked as active (`status` = 'Current') but have generated less than $10,000 in total revenue or sold fewer than 50 total units.
- **Tables & Columns:**
  - `factinternetsales`: `salesamount`, `orderquantity`, `productkey`.
  - `product`: `productkey`, `productname`, `status`.
  - `calendar`: `datekey`, `calendaryear`.
- **Technical Hint:** Apply a `where` clause to filter for the current status and the specific year. Use a `having` clause after your `group by` to strictly filter out products that exceed your minimum revenue and unit thresholds.

15. How does product pricing tier affect consumer purchasing behavior?

- **Business Context:** Do we sell predominantly budget-friendly items or premium gear? Grouping products by price tier helps merchandising teams understand the primary customer budget.
- **The Task:** Categorize products into three price tiers based on their `listprice` (e.g., Entry-Level: < $100, Mid-Range: $100 - $500, Premium: > $500). Calculate the total revenue and total units sold for each tier.
- **Tables & Columns:**
  - `factinternetsales`: `salesamount`, `orderquantity`, `productkey`.
  - `product`: `productkey`, `listprice`.
- **Technical Hint:** Use a `case when` statement evaluating the `listprice` to define your Entry-Level, Mid-Range, and Premium labels. Then, `group by` this `case` statement and aggregate your sales and quantity metrics.

Section 4: Promotional ROI & Sales Drivers

16. What are the primary psychological drivers (Sales Reasons) behind our B2C purchases?

- **Business Context:** Marketing needs to know what messaging works. Are customers buying because of "Price", "On Promotion", or "Manufacturer" reputation? This directly dictates future ad copy.
- **The Task:** Calculate the total revenue and the total number of distinct orders associated with each `salesreasonname` and `salesreasonreasontype`.
- **Tables & Columns:**
  - `factinternetsales`: `salesordernumber`, `salesamount`.
  - `factinternetsalesreason`: `salesordernumber`, `salesreasonkey`.
  - `salesreason`: `salesreasonkey`, `salesreasonname`, `salesreasonreasontype`.
- **Technical Hint:** You must use an `inner join` from the main fact table to the bridge table (`factinternetsalesreason`), and then another `inner join` to the `salesreason` dimension. Use `sum` for revenue and `count distinct` for the orders, then `group by` the reason name and type.

17. What percentage of our total B2C volume is driven by promotional discounts versus standard pricing?

- **Business Context:** If a company relies too heavily on discounts, it trains customers to never buy at full price, destroying long-term brand equity. We need to measure our discount dependency.
- **The Task:** Segment all historical orders into two buckets: "Discounted" and "Full Price". Calculate the percentage of total orders that fall into each bucket.
- **Tables & Columns:**
  - `factinternetsales`: `salesordernumber`, `promotionkey`.
  - `promotion`: `promotionkey`, `discountpct`.
- **Technical Hint:** In AdventureWorks, a `promotionkey` of 1 typically means "No Discount". You can use a `case when` statement to flag rows where the promotion key is not 1 (or `discountpct` > 0) as 'Promotional' and the rest as 'Standard'. Then calculate the distribution.

18. Does offering a promotional discount actually increase the average basket size (quantity), or does it just cannibalize our margins?

- **Business Context:** The only mathematical justification for lowering a price is if the customer buys *more* units to offset the lost margin. If they buy the exact same amount but pay less, the promotion is a failure.
- **The Task:** Compare the average `orderquantity` and the average Gross Margin percentage for items sold under a promotion versus items sold at full price.
- **Tables & Columns:**
  - `factinternetsales`: `orderquantity`, `salesamount`, `totalproductcost`, `promotionkey`.
- **Technical Hint:** Use the same `case when` logic from Question 17 to group the data into 'Promotional' and 'Standard' buckets. Use the `avg` function on the `orderquantity` and calculate the gross margin.

19. Do the primary sales drivers change depending on the Product Category?

- **Business Context:** Customers might buy a $2,000 bicycle because of "Quality" or "Manufacturer", but they might buy a $10 water bottle purely because of "Price". Knowing this allows for hyper-targeted marketing per category.
- **The Task:** For each `productcategoryname`, identify the single most frequent `salesreasonname` (the reason associated with the highest number of orders).
- **Tables & Columns:**
  - `factinternetsales`: `salesordernumber`, `productkey`.
  - `factinternetsalesreason`: `salesordernumber`, `salesreasonkey`.
  - `salesreason`: `salesreasonname`.
  - `product`: `productkey`, `productsubcategorykey`.
  - `productsubcategory`: `productsubcategorykey`, `productcategorykey`.
  - `productcategory`: `productcategoryname`.
- **Technical Hint:** This is a fantastic place to showcase advanced SQL. First, create a `cte` that joins all these tables, groups by category and reason, and counts the distinct orders. Inside that same `cte`, use the `row_number()` function `over` a `partition by` the category name, and `order by` the order count `desc`. Finally, write a main query that selects `from` this `cte` `where` the row number equals 1.

Section 5: Geographic Distribution & Fulfillment Efficiency

20. Which countries and regions drive the most B2C revenue, and how does the Average Order Value (AOV) differ geographically?

- **Business Context:** The marketing team needs to know where to allocate their regional ad spend. If Australia has fewer orders but a significantly higher AOV than the UK, the ROI on an Australian ad campaign might be much better.
- **The Task:** Calculate the total revenue, total number of distinct orders, and the Average Order Value for each `salesterritorycountry` and `salesterritoryregion`.
- **Tables & Columns:**
  - `factinternetsales`: `salesamount`, `salesordernumber`, `salesterritorykey`.
  - `salesterritory`: `salesterritorykey`, `salesterritorycountry`, `salesterritoryregion`.
- **Technical Hint:** Simply use an `inner join` between the fact table and the territory dimension. Then, `group by` the country and region. Use `sum` for the revenue and divide it by the `count distinct` of the order numbers to find your AOV.

21. Are we meeting our operational Service Level Agreements (SLAs)? What is the average lead time between Order Date and Ship Date?

- **Business Context:** In the age of Amazon Prime, slow shipping kills repeat business. The operations team needs to know exactly how many days it takes for the warehouse to get a product out the door.
- **The Task:** Calculate the average number of days it takes to ship an order after it is placed, and identify if any specific months or regions have significant shipping delays.
- **Tables & Columns:**
  - `factinternetsales`: `orderdate`, `shipdate`, `salesterritorykey`.

  - `salesterritory`: `salesterritoryregion`.
  - `calendar`: `monthname` (linked via `orderdatekey`).
- **Technical Hint:** You will need the `datediff` function in sql server to find the integer difference in days between the `orderdate` and `shipdate`. Wrap this inside an `avg` function. You can then `group by` the region and month to look for bottlenecks.

22. How heavily do freight and tax costs impact the final landed cost for the consumer across different countries?

- **Business Context:** If freight costs to Europe are eating up 20% of the total order value, the company might need to negotiate new logistics contracts or open a local distribution center.
- **The Task:** Calculate the total `freight` and total `taxamt` as a percentage of the total `salesamount` for each country.
- **Tables & Columns:**
  - `factinternetsales`: `salesamount`, `freight`, `taxamt`, `salesterritorykey`.
  - `salesterritory`: `salesterritorycountry`.
- **Technical Hint:** Use `sum` on all three financial metrics, grouped by the country. Then, divide the sum of freight by the sum of sales to get your freight ratio. Formatting these outputs cleanly will look fantastic when you document the query results in Obsidian or AsciiDoc later.

23. Is there a regional bias for specific Product Categories?

- **Business Context:** Do North American customers buy more mountain bikes while European customers buy more road bikes? Knowing this helps supply chain planners stock the correct regional warehouses.
- **The Task:** For each `salesterritorycountry`, identify the total revenue generated by each `productcategoryname`. Calculate what percentage of that country's total revenue comes from each category.
- **Tables & Columns:**
  - `factinternetsales`: `salesamount`, `salesterritorykey`, `productkey`.
  - `salesterritory`: `salesterritorycountry`.
  - `product`: `productkey`, `productsubcategorykey`.
  - `productsubcategory`: `productsubcategorykey`, `productcategorykey`.
  - `productcategory`: `productcategoryname`.
- **Technical Hint:** You will need to join several tables together here. To get the percentage, you can use a window function like `sum over ( partition by salesterritorycountry )` to find the denominator (total country sales), and divide the specific category sales by that denominator. This is a perfect query to visualize later in Tableau as a 100% stacked bar chart.